$$L = -\left[y\log(a) + (1-y)\log(1-a) \right]$$

$$\frac{dL}{da} = -\left[\frac{y}{a} + (1-y)\cdot\frac{1}{1-a}\cdot(0-1) \right]$$

$$= -\left[\frac{y}{a} + \frac{1-y}{1-a}\cdot(-1) \right]$$

$$= -\left[\frac{y}{a} - \frac{1-y}{1-a} \right]$$

$$= -\frac{y}{a} + \frac{1-y}{1-a}$$

$$= \frac{-y(1-a) + a(1-y)}{a(1-a)}$$

$$= \frac{-y + \cancel{ay} + a - \cancel{ay}}{a(1-a)}$$

$$\therefore \frac{dL}{da} = \frac{a-y}{a(1-a)}$$